

Synopsis: Spontaneous $\mathbb{Z}_2$ Symmetry Breaking via Hysterical Instability

Master theorem and a catalogue of odd channels

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

The paper proves a channel-agnostic branch-asymmetry theorem and catalogues interesting $\mathbb{Z}_2$ portals. It does not claim Hysterical causation for any observed asymmetry. Weak parity as a branch-selected remnant of a symmetric parent Lagrangian is speculation only.

1 Master theorem

If a supercritical odd Hysterical mode Y couples through an odd neutral portal to an activity kernel $K \geq 0$, and washout-weighted transport Φ is odd and strictly positive on nontrivial nonnegative sources, then on opposite histories

$$\eta[Y_+] = -\eta[Y_-] \neq 0$$

whenever overlap ($K > 0$ and $\dot{Y} \neq 0$ concurrently) holds. The $W\widetilde{W}$ and $R\widetilde{R}$ channels are named instances of this theorem.

2 Catalogue

A five-column ledger lists symmetries with odd coordinate, neutral portal, early-universe activity, and whether Nature shows a broken yield. Highlights: strong CP and dark $\mathbb{Z}_2$ have portals without broken yields; baryon/CP do show broken yields; weak parity is already chiral in the SM Lagrangian.

3 Weak-parity speculation

Optionally, a left-right / mirror-symmetric parent Lagrangian could be reduced to the observed chiral SM by Hysterical selection of one hand. That is an interpretive gloss, not a theorem.

4 Conclusion

$$\text{odd supercritical } Y \xrightarrow{\text{portal} + \text{overlap}} \eta[Y_+] = -\eta[Y_-] \neq 0.$$